Open to internships & entry-level roles

Bahesty Monfared

Engineer in the making, building embedded IoT systems



Final-year BSc student in Electrical and Electronic Engineering at **AIUB**, graduating December 2026. I lead **HydroSense**, a smart drip-irrigation system for terraced farms in Bangladesh — I designed the system, wrote the ESP32 firmware, and built its bilingual app and website. Before that, I spent years teaching, which is where I learned that a design is only finished when somebody who wasn't in the room can follow it.

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3.38 / 4.00	100+	50+	9	Dec 2026
CGPA, BSC EEE	STUDENTS TAUGHT	VOLUNTEERS TRAINED	PROJECTS ON THIS PAGE	GRADUATION

Featured project

Final-year capstone · AIUB EEE · Group 11

WEBSITE LIVE · HARDWARE BUILD IN PROGRESS · SUPERVISOR: ASSOC. PROF. RETHWAN FAIZ

HydroSense — smart drip irrigation for terraced farms

Every drop, right where the roots need it.

On sloped and terraced land, water runs off before it can soak in. HydroSense reads soil moisture in three separate zones, checks the weather forecast, and waters the driest zone first in short pulses — so the water goes into the soil instead of down the hill. The ESP32 makes every watering decision itself, so the farm keeps working when the internet drops; the cloud only gives advice.

Team Leader | App Developer | System Designer

- **Led the four-member student team** and designed the system architecture the rest of the build follows.
- **Wrote the ESP32 firmware** (Arduino C++) — a 60-second control loop, driest-zone-first scheduling, and pulse and volume caps that keep water below the soil's soak-in rate.
- **Built the bilingual app** in English and বাংলা with Capacitor, synced through Firebase and shared as an Android APK.
- **Produced the hardware design pack** — a Rev C bill of materials, an 8-page circuit diagram covering 7 subsystems, a connection manual, and an interconnection design that fixed a 5 V / 3.3 V I²C clash with a BSS138 level shifter.
- **Built the project website** with a live decision-logic console, and wrote the engineering write-up and capstone book chapters.

ESP32 · ADS1115 16-bit ADC · Capacitive moisture probes · Hall-effect flow sensors · DHT22 · MOSFET PWM + relay · Firebase · Open-Meteo API · Capacitor · FAO-56

Sense

Capacitive probes in each of the three zones are read through a 16-bit ADS1115 ADC, because the ESP32's own ADC2 can't be used while Wi-Fi is on. A DHT22 tracks temperature and humidity, and hall-effect flow sensors confirm how much water was actually delivered.

Decide

Every 60 seconds the ESP32 checks each zone against its dry threshold and looks at the live Open-Meteo forecast. If rain is on the way, it skips watering. If not, the driest zone goes first.

Water

A 12 V diaphragm pump — switched by a MOSFET with a relay as a hardware enable — feeds pressure-compensating drip emitters. Short pulses with volume caps stop the water from running off the slope.

Sync & learn

Readings sync to Firebase over a 4G router, and a 5-day FAO-56 water plan tunes itself to the field in about two weeks. The plan only advises: the ESP32 always applies the smaller of its own limit and the plan's target.

3	60 s	5-day	2	10 yr	≈71%*
INDEPENDENT ZONES	CONTROL CYCLE	WATER FORECAST	LANGUAGES	ERA5 CLIMATE DATA	LESS WATER

* From FAO-56 simulation runs. Bench and field tests are still under way, so this is not a measured result yet.

Recent work

Latest first · Jul–Nov 2026

05 Nov 2026 — PulseGuard proposal submitted

A group project on embedded machine learning for physiological monitoring. Our group started on 15 October; the proposal went in on 5 November, and the build comes next.

Nov 2026 — HydroSense moves to 4G (Rev E)

The SIM800L 2G module couldn't hold a secure (TLS) connection to Firebase, so the design switched to a 4G pocket router powered from the 5 V rail.

31 Oct 2026 — HydroSense hardware build begins

Started putting the prototype together by testing each component on its own before connecting the full system.

Oct 2026 — HydroSense Rev C hardware design pack

Bill of materials, an 8-page circuit diagram covering 7 subsystems, a 4-page connection manual, and an interconnection design that solved a 5 V / 3.3 V I²C bus clash with a BSS138 level shifter.

Oct 2026 — Predictive water-demand engine

A 5-day FAO-56 water plan for every zone that tunes itself to the field, plus a yearly demand curve from 10 years of ERA5 climate data — with a standalone demo and CSV export.

Sep 2026 — ParkTag smart parking designed

An Arduino and NFC parking system with automatic billing in taka — proposal, full wiring diagram, and a costed component list.

Aug 2026 — Solar–grid project book and IEEE paper

The project book cleared iThenticate at 8% similarity with no integrity flags. We then condensed it into a 6-page IEEE-format paper and prepared the final defence.

Jul 2026 — Solar–grid prototype tested

Recorded prototype data from 19 to 27 July. The team's diagnostics dataset reached 302,915 samples — 301,696 after cleaning.

Projects

Filter, or pick a skill below to see where I used it

CAPSTONE · TEAM LEADER — ● BUILDING

HydroSense smart drip irrigation

Three-zone, weather-aware drip irrigation for terraced farms, with a bilingual app and a live project website.

AIUB EEE · Group 11 · 2026

ESP32 · Arduino C++ · Firebase · Capacitor · Simulink

HYDROSENSE · SOFTWARE MODULE — ● IMPLEMENTED

Predictive water-demand engine

Plans irrigation five days ahead and learns the field it is installed in — without ever taking control away from the controller.

JavaScript · pure engine, no DOM or Firebase dependencies

JavaScript · FAO-56 · Open-Meteo · ERA5

- **Layer 1:** a 5-day FAO-56 water balance for each zone that produces an advisory watering plan.
- **Layer 2:** learns three bounded factors from soil-probe history — crop coefficient Kc, VPD sensitivity and the ET_0 blend weight — starting from FAO-56 book values and settling in about 14 clean days.
- **Layer 3:** a day-of-year demand curve built from 10 years of ERA5 climate data.
- **Safe by design:** the ESP32 applies min(local cap, plan target), discards plans older than 36 hours, and the plan can never loosen local limits.
- Ships with a standalone demo page that fetches the live Open-Meteo forecast and exports CSV.

TEAM PROJECT · GROUP 2026.2.7 · PROTOTYPE TESTED

Hybrid solar–grid fast charger with AI diagnostics

Solar first, grid as backup: fast charging with MPPT, smart battery protection and a machine-learning model that spots faults from live sensor data. An ESP32 edge node reports to a Raspberry Pi Zero 2 W server.

AIUB · four members · prototype data 19–27 Jul 2026

ESP32 · Raspberry Pi Zero 2 W · MPPT · BMS · Python ML

- **My parts:** environmental impact, and project finance and costing — component cost BDT 41,712.
- **Simulation vs hardware:** compared simulated behaviour with the real prototype, and presented the live demonstration.
- **Writing:** wrote chapters of the six-chapter project book (8% iThenticate similarity, no integrity flags) and co-wrote a condensed 6-page IEEE-format paper.
- **Team dataset:** 302,915 raw samples, 301,696 after cleaning.

GROUP PROJECT · PROPOSAL SUBMITTED

PulseGuard — embedded ML for physiological monitoring

A group project that brings machine learning onto embedded hardware to monitor physiological signals. Work started on 15 October 2026 and the proposal was submitted on 5 November 2026.

AIUB · group · Oct 2026 – present

Embedded systems · Machine learning · Physiological sensing

COURSE PROJECT · DESIGNED

ParkTag — NFC smart parking system

An Arduino parking system where an NFC card is both ticket and key: tap in and the gate opens; tap out and the screen shows your time and bill in taka.

Arduino Uno R3 · Dhaka-sourced parts · Sep 2026

Arduino Uno · MFRC522 NFC/RFID · DS3231 RTC · I²C LCD · Servo gates

- Two NFC/RFID readers (entry and exit), four IR slot sensors, two servo barrier gates, and a 16×2 I²C screen at each gate.
- Billing in BDT with a first-hour minimum, an hourly rate, a daily cap and a grace period; a battery-backed DS3231 clock keeps billing right through power cuts.
- Picked IR over ultrasonic sensors to save pins, and an I²C backpack to cut each display from six pins to two.
- Added an AMS1117 3.3 V regulator so two RFID readers don't overload the Uno; a red LED and buzzer warn when all four slots are full.
- Delivered a project proposal, a full wiring diagram, and a component list costed in taka with a contingency budget.

COURSE PROJECT · RENEWABLE ENERGY TECH. · COMPLETED

Off-grid solar PV design for Bangladesh

Sizing and simulation of a standalone solar system under Bangladeshi sunlight and load conditions, ending in a 2S × 4P eight-module array.

AIUB · group · IEEE-format report

MATLAB · Simulink · PV sizing · IEEE writing

- Modelled the array in MATLAB and Simulink and produced the I–V and P–V curves used to justify the configuration.
- Worked the sizing by hand first — sunlight, module output, series and parallel layout — so the simulation checked the reasoning instead of replacing it.
- Wrote it up in IEEE format, stating every assumption and citing a source for every figure that wasn't measured.

UNDERGRADUATE THESIS · IN PROGRESS

How colour and layout steer attention on web pages

A group study of how visual design choices change where people actually look — evaluating eye-tracking datasets, then designing our own tests for the gaps.

AIUB · group research

Research methods · Dataset evaluation · Experiment design

- Compared published eye-tracking and saliency datasets — UEyes, Imp1k, FiWI — against what the study needed.
- Helped design tests for the gaps: A/B tests, five-second memory tests, blur tests, distraction tests and side-by-side ranking.

COURSEWORK · HCI · COMPLETED

Usability analysis of the Steam desktop app

A screen-by-screen check of a large desktop app against Don Norman's design principles, delivered as an annotated presentation.

AIUB · sole author

Usability heuristics · Annotated screenshots

- Judged a messy real interface against a fixed checklist — visibility, feedback, affordance, mapping, constraints, consistency — instead of giving a general opinion.
- Marked the evidence directly on screenshots so a reader can check each conclusion for themselves.

TEACHING MATERIAL · COMPLETED

Animated HTML & web design lessons

A lesson set for the HSC ICT web design unit, with the demos coded rather than described — students watch the markup and the result change together.

HSC ICT curriculum · built from scratch

HTML · CSS · PowerPoint animation

Experience

Teaching · training · customer-facing work

ACADEMIC TEACHER

Carupath Academic Coaching
IT department
Jan 2024 — Sep 2025

Teaching IT to students starting from zero

100+ students taught **1 yr 9 mo** in the role

- **Wrote step-by-step practical exercise sheets** and lesson material for mixed-level classes.
- **Kept marks and attendance** in Excel and Google Sheets, and prepared progress reports and result sheets for parents and faculty — keeping student information confidential.
- **Stayed in touch with parents and students** by phone and WhatsApp, and helped bring in new enrolments.
- **Handled day-to-day admin** — scheduling sessions, sending reminders, drafting notices, and running exam days.

BRAND PROMOTER

ACI Limited
Twinkle Baby Pant Diaper
Feb 2023 — Nov 2023

Face-to-face sales and product communication

50+ outlets & events **100+** customers on a typical day

- **Demonstrated product features and made sales**, meeting my sales target most of the time.
- **Checked stock and displays** against brand guidelines across retail outlets and events.

VOLUNTEER TEACHER

Bidyanondo Foundation
Jan 2022 — Dec 2023

Teaching and training volunteers in a resource-limited setting

50+ volunteers onboarded & trained

- **Taught with limited materials** and a wide spread of ability in one room, rebuilding the same explanation until it landed for everybody.
- **Helped screen and recruit new volunteers**, then trained them through group sessions, one-to-one guidance and shared notes.
- **Helped run programmes and events** for the foundation.

Education

All BSc coursework complete · capstone & thesis remaining

BSC — FINAL YEAR

Expected December 2026
CGPA 3.38 / 4.00

Electrical & Electronic Engineering

American International University–Bangladesh (AIUB), Dhaka

Digital Logic & Circuits (Verilog) · Control Systems · Renewable Energy Technology · Nuclear Power Engineering · Numerical Methods (MATLAB) · Power Systems · Web Technologies · Human–Computer Interaction · Principles of Accounting · Principles of Economics · Engineering Economics · Industrial Management · Human Resource Management · Organizational Behaviour · Business Communication

HSC — SCIENCE

2021 · Dhaka Board
GPA 5.00 / 5.00

Mazid Jarina Foundation School and College

SSC — SCIENCE

Shaheed Police Smrity College, Dhaka

2019 · Dhaka Board
GPA 4.39 / 5.00

Skills

Tools I have actually built something with

EMBEDDED & HARDWARE

- ESP32
- Arduino Uno
- Raspberry Pi
- Sensor integration
- I²C & level shifting
- NFC / RFID
- Relay & MOSFET pump control

PROGRAMMING & APPS

- C / C++
- JavaScript
- HTML / CSS
- Python
- Firebase
- Capacitor (Android)
- Verilog

DESIGN & SIMULATION

- MATLAB
- Simulink
- AutoCAD
- NI Multisim
- PSpice
- Proteus

ENERGY & WATER

- PV system sizing
- MPPT
- BMS
- FAO-56 water balance
- Project costing (BDT)

WRITING & RESEARCH

- IEEE technical writing
- Research methods
- Usability evaluation
- Project books
- MS Word / Excel / PowerPoint
- Google Workspace

PEOPLE & TEACHING

- Lesson design
- Volunteer training
- LMS (Google Classroom)
- Parent & client communication
- Sales & customer service

LANGUAGES

- Bangla — native
- English — professional working proficiency

Get in touch

Based in Dhaka · on-site, hybrid or remote

I’m looking for internships and entry-level roles where I can keep working on real hardware and real documentation — and on roles where explaining technical work clearly to people outside the team matters.

- Embedded & IoT
- Power & renewable energy
- Application engineering
- Operations & supply chain
- HR, training & L&D
- Business development

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> **One thing stated plainly:** I am in my final year and graduate in December 2026 — I don’t hold the degree yet. All my coursework is complete; the capstone and thesis remain.
>
> I’d rather say that here than have it come up later. Everything on this page is work I have actually done.

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